BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 8 — Geometry of Straight Lines

Total Marks: 65	**Time Allowed: 2h 30m**	**Version: 1**

General Instructions:

1. Write your name, roll number, and section clearly on the answer sheet.
2. **Section A** (MCQs): Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
3. **Section B** (Short Questions): Attempt **ALL 11** pairs, choosing the main question **OR** its alternative. Each carries **3 marks**.
4. **Section C** (Long Questions): Attempt **ALL 4** pairs, choosing the main question **OR** its alternative. Each carries **5 marks**.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	The gradient of a straight line is defined as:	rise ÷ run	run ÷ rise	rise × run	rise + run	◯◯◯◯
2	The gradient of a horizontal line is:	1	undefined	0	−1	◯◯◯◯
3	If m_1 and m_2 are gradients of two perpendicular lines, then:	$m_1 = m_2$	$m_1 + m_2 = 0$	$m_1 \times m_2 = 1$	$m_1 \times m_2 = -1$	◯◯◯◯
4	Slope-intercept form of a line with slope m and y-intercept c is:	$y = mx + c$	$y - y_1 = m(x - x_1)$	$x/a + y/b = 1$	$ax + by + c = 0$	◯◯◯◯
5	The slope of the line $6x - 5y + 15 = 0$ is:	6/5	−6/5	5/6	−5/6	◯◯◯◯
6	Equation of a line parallel to the y-axis through (3, 5) is:	$y = 5$	$x = 3$	$y = 3$	$x = 5$	◯◯◯◯
7	The x-intercept of the line $x + y = 5$ is:	5	−5	1	−1	◯◯◯◯
8	For two lines with slopes m_1, m_2: $\tan\theta =$	$(m_2-m_1) / (1+m_1m_2)$	$(m_2+m_1) / (1-m_1m_2)$	$m_1m_2 / (1+m_1+m_2)$	$(m_2-m_1) / (1-m_1m_2)$	◯◯◯◯
9	Three points A, B, C are collinear if:	Slope of AB ≠ slope of BC	Slope of AB = slope of BC	AB + BC = AC	AB ⊥ BC	◯◯◯◯
10	In normal form $x \cos\alpha + y \sin\alpha = p$, p represents:	Slope of the line	The y-intercept	Length of perpendicular from origin	The x-intercept	◯◯◯◯
11	The line $y = 0$ represents:	y-axis	x-axis	a point	a plane	◯◯◯◯
12	Point of intersection of $9x - 7y = 0$ and $8x - 11y = 0$ is:	(0, 0)	(0, 7)	(9, 0)	(7, 9)	◯◯◯◯

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt ALL 11 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	Find gradient of line with inclination 120°. State whether positive or negative and why.	**3**	**OR**	Find inclination of a line whose slope is −1.732.	**3**
2(ii)	Find the gradient and inclination of the line joining A(2, 6) and B(5, 8).	**3**	**OR**	Find gradient and inclination of the line joining C(−2, 4) and D(1, −3).	**3**
2(iii)	A(−2, 6), B(7, −3): find slope of line (a) parallel to AB, (b) perpendicular to AB.	**3**	**OR**	Find x if slope of line through A(3, x) and B(5, 8) is 4.	**3**
2(iv)	Using slopes, prove X(0, −3), Y(4, 7) and Z(6, 12) are collinear.	**3**	**OR**	Find y if points P(4, y), Q(5, 2) and R(6, 2y+1) are collinear.	**3**
2(v)	Find equation of line with slope = 2 and y-intercept = −3. Write in general form.	**3**	**OR**	Find equation of line through (−5, 7) with slope 4 using point-slope form.	**3**
2(vi)	Find equation of line through (2, 4) and (−1, −4) using two-point form.	**3**	**OR**	Find equation of line with x-intercept = −6 and y-intercept = 5 using two-intercept form.	**3**
2(vii)	Reduce $6x - 5y + 15 = 0$ to slope-intercept form. State slope and y-intercept.	**3**	**OR**	Reduce $4x - 3y + 9 = 0$ to two-intercept form. State x-intercept and y-intercept.	**3**

2(viii)	Find equation of line through (3, −2) and perpendicular to a line with slope 3.	**3**	**OR**	Find equation of line with y-intercept = −7 and parallel to a line with slope 4.	**3**
2(ix)	Find angle from l_1 to l_2 if slope of $l_1 = -1$ and slope of $l_2 = 2$.	**3**	**OR**	Find angle between lines 3x + 2y + 5 = 0 and 2x − 3y + 8 = 0.	**3**
2(x)	Find point of intersection: 2x + y + 1 = 0 and x − y − 4 = 0.	**3**	**OR**	Find point of intersection: x + y + 3 = 0 and 2x − 5y + 8 = 0.	**3**
2(xi)	Find equation of line through intersection of x + 2y = 0, x − y = 3 and point (1, 1).	**3**	**OR**	Find equation of family of lines through intersection of 6x + 5y + 3 = 0 and 2x − 5y + 13 = 0 with slope 3.	**3**

SECTION C — Long Questions (4 × 5 = 20 Marks)

Attempt ALL 4 questions. Choose either the main question OR its alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Using slopes, prove A(3, −1), B(−5, −5), C(1, 3) are vertices of a right-angled triangle. Identify the right angle and state slopes of all three sides.	**5**	**OR**	Using slopes, prove A(−2, 1), B(6, 3), C(10, 5), D(2, 3) are vertices of a parallelogram. Show both pairs of opposite sides are parallel.	**5**
3(ii)	Reduce 6x − 5y + 15 = 0 into ALL six standard forms: (i) slope-intercept, (ii) point-slope, (iii) two-point, (iv) two-intercept, (v) normal form, (vi) symmetric form.	**5**	**OR**	Find equation of perpendicular bisector of segment joining A(0, 6) and B(2, −2). Also find where this bisector meets the x-axis and y-axis.	**5**
3(iii)	Find all three interior angles of triangle ABC with A(−2, 0), B(3, 0), C(6, 5). Use $\tan\theta = (m_2-m_1)/(1+m_1m_2)$ for each angle. Verify sum = 180°.	**5**	**OR**	Find all three interior angles of triangle XYZ with X(−2, 3), Y(−3, −4), Z(5, 2). Use the angle formula for each vertex.	**5**
3(iv)	A cricket team scores 96 runs in 16 overs and 180 runs in 30 overs. (i) Write equation using two-point form. (ii) What does gradient mean in terms of scores? (iii) Predicted score after 45 overs? (iv) After how many overs will score be 240?	**5**	**OR**	Nasir sold fruiters at Rs. 120 per dozen and Shakri Malta at Rs. 150 per dozen, earning Rs. 1200. (i) Write equation in standard form. (ii) If he sold 4 dozen Shakri Malta, find dozens of fruiters sold. (iii) Write in slope-intercept form and state what slope means.	**5**

Invigilator's Signature: ____________________ Candidate's Signature: ____________________

Chapters Covered: Ch 8 — Geometry of Straight Lines

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